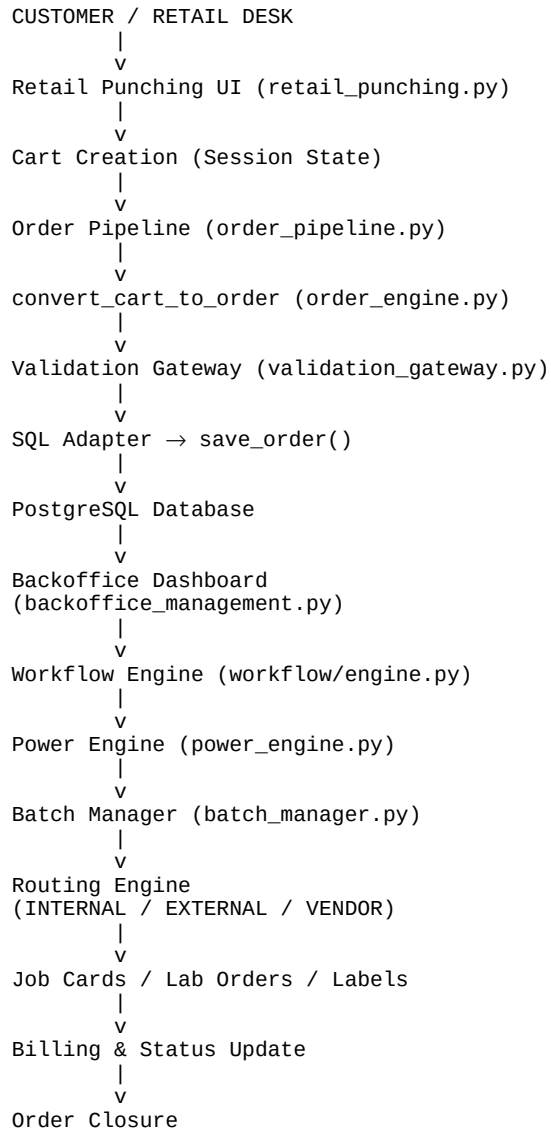


Order Processing & Workflow Architecture

This document explains how an order is punched, processed, routed, and completed using various modules in the WIN07 system.

System Flowchart



Module Responsibilities

1. retail_punching.py
 - Captures RX, product, quantity, patient data
 - Builds cart
2. order_pipeline.py
 - Converts cart to order
 - Triggers validation
 - Saves order
3. order_engine.py

- Builds standardized order structure

4. validation_gateway.py

- Credit, fraud, pricing validation

5. sql_adapter.py

- Database CRUD operations
- fetch_orders, save_order, fetch_full_order

6. backoffice_management.py

- Main workflow UI
- Allocation, pricing, job cards

7. power_engine.py

- Vertex correction
- Manufacturing power

8. batch_manager.py

- FIFO allocation
- Stock lookup

9. workflow/engine.py

- Orchestrates complete workflow
- Routing + status

10. workflow/status.py

- Order status lifecycle

Data Flow Summary

Retail UI → Session → Order Dict → Database → Backoffice UI
→ Workflow Engine → Production / Vendor → Billing

Status Lifecycle

PENDING
↓
CONFIRMED
↓
ALLOCATED
↓
IN_PRODUCTION
↓
READY
↓
DISPATCHED
↓
DELIVERED

Future Expansion Plan

- Vendor API integration
- Auto PO generation
- Lab SLA monitoring
- Real-time stock sync
- Analytics dashboard

Conclusion

This workflow ensures complete traceability from punching to delivery. The modular architecture allows independent scaling of retail, lab, and vendor operations.